JY-CRPG-BENCH: BENCHMARKING LONG-HORIZON AGENCY IN AN OUT-OF-DISTRIBUTION OPEN WORLD

Han Xiao
Jina AI
han.xiao@jina.ai

ABSTRACT

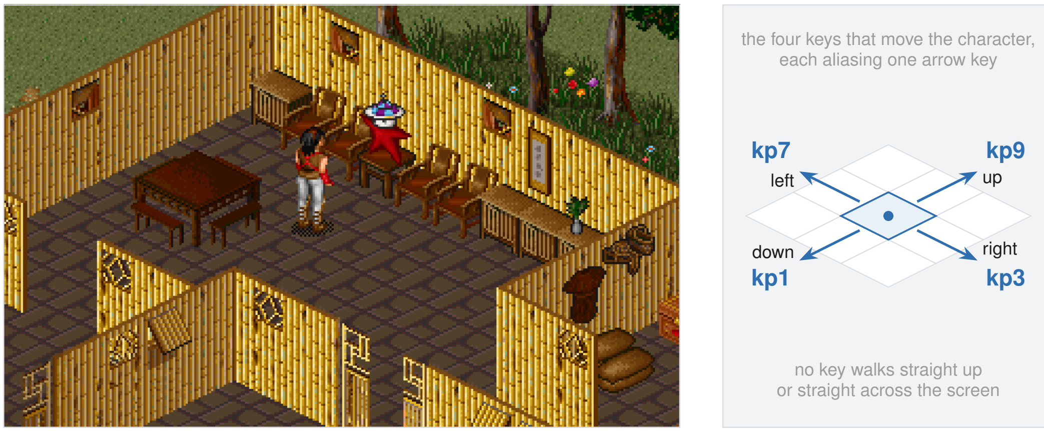
Long-horizon agency requires an agent to perceive an unfamiliar world, decide without being told what is worth doing, and stay coherent over hundreds of dependent decisions. We introduce JY-CRPG-BENCH, a benchmark that places a vision-language agent inside 金庸群俠傳 (*Heroes of Jin Yong*, Heluo Studio 1996), an unmodified commercial open-world role-playing game run under DOS emulation. The environment is deliberately far from the distribution modern vision encoders are trained on: a 320×200 framebuffer, a $45°$ isometric projection in which no key moves straight up the screen, Traditional Chinese rendered as 16-pixel bitmap glyphs, and a world whose objectives are stated only in its own fiction. An agent joins by pointing an HTTP client at a session URL, so any model under any harness can play without adopting a framework. We score progress from the state of the emulated machine: character records, milestone events and scene transitions are read out of memory, so each rung of progress is verifiable and cannot be reached by pixel similarity. Across 13 agent configurations from 6 vendors and a seeded random baseline, no session gains a single experience point or passes level one within the twenty-minute budget, and 11 of 19 sessions never leave the scene they start in. Behaviour nevertheless separates sharply beneath that floor: the screen-changing action ratio spans 0.068–0.898 against a random-policy floor of 0.403, and one configuration falls below the floor while reading the screen before nearly every action. JY-CRPG-BENCH is released as a live, replayable public arena at https://github.com/hanxiao/jy-crpg-bench.

1 INTRODUCTION

Frontier models prove competition mathematics and write working compilers, yet remain unreliable on tasks that unfold over hours. Surveys of this gap converge on a single diagnosis: outcome-only signals become uninformative as horizons lengthen, so evaluation has to manufacture denser step-level evidence to say anything at all (Chen et al., 2026). Long-horizon benchmarks answer this in two families. Episodic benchmarks set a goal and score its completion. Continuing benchmarks give the agent an environment with no terminal state and score what accumulates (Backlund & Petersson, 2025; Fan et al., 2026). Open-world games belong to the second family and add a requirement the business simulations do not: the agent must recover its own state from raw perception, because nothing in the environment reports it.

Existing game benchmarks each remove one part of that problem to isolate another. BALROG evaluates six research environments in language-only and language-vision modes (Paglieri et al., 2025); TextQuests removes perception entirely (Phan et al., 2025); ARC-AGI-3 excludes language and cultural priors by construction (ARC Prize Foundation, 2026); VideoGameBench detects progress in real-time action games by matching against checkpoint images taken from walkthroughs (Zhang et al., 2025); MineExplorer filters out tasks whose solutions depend on game-specific knowledge (Ju et al., 2026). Each removal is principled, and each leaves the combination untested. We keep the parts together, because acting through unfamiliar perception, unfamiliar language and unfamiliar culture at once is what a deployed agent is asked to do.

JY-CRPG-BENCH runs one deep environment in place of a wide suite, following the precedent set by Crafter and NetHack (Hafner, 2021; Küttler et al., 2020). The environment is 金庸群俠傳,



(a) the frame every session boots into, 320×200, upscaled without interpolation **(b)**

Figure 1: The observation and the control geometry. Panel (a) is the opening scene, with the protagonist centre-left and the first speaking character to the right. Panel (b) shows why screen-space reasoning fails: the four movement keys run along the diagonals of the projection.

a 1996 Chinese role-playing game that remains a landmark of its genre, executed as its original DOS binary under emulation with no modification. The game presents an open world built from the fourteen novels of Jin Yong: a two-tier map, several hundred named characters each carrying level, experience, skills, items and reputation, and a completion condition that requires recovering fourteen books scattered across the world. A first human playthrough takes tens of hours. The agent receives the framebuffer and a key vocabulary, and nothing else.

Three properties make the environment adversarial for current models in ways a modern title would not be, and Figure 1 shows the first two of them in the frame every session starts from. Perception is out of distribution on several axes at once: native-resolution studies report that the usual downsampling path costs vision-language models accuracy (Niu et al., 2025), corruption studies measure systematic sensitivity to degraded input (Wasim et al., 2025), out-of-distribution image benchmarks find degradation even on familiar categories (Lin et al., 2026), and text rendered as pixels is understood less well than the same content supplied as text (Liu et al., 2026). A 320×200 frame of 1996 bitmap art with 16-pixel Traditional Chinese glyphs sits at the hard end of all four. Control is projected: the world is drawn in a 45° isometric projection, so the four movement keys run along screen diagonals and an agent reasoning in screen coordinates walks in circles. Progress is self-directed: the game keeps no quest log, draws no objective marker, and reports no score, so deciding that the chest is worth searching and that the compass gates the rest of the world is the problem the agent has to solve. ARC-AGI-3 places the same goal-setting requirement at the centre of agency (ARC Prize Foundation, 2026).

Measuring such an environment from the screen is unreliable, and the field has begun to say so: GameWorld names heuristic verification as the obstacle to standardized game-agent evaluation and pairs each of its tasks with a state-verifiable metric (Ouyang et al., 2026). We take the same position and push it into the emulator. The benchmark reads progress out of the memory of the running machine: the character array the game itself maintains, at 320 records of 182 bytes, and the fade-to-black the renderer performs on every scene change. The agent never sees any of this. Screen-derived quantities are reported as behaviour, never as progress, so no amount of visual churn can move a progress metric.

We make three contributions.

1. We present JY-CRPG-BENCH, a long-horizon benchmark on an unmodified commercial open-world role-playing game, whose difficulty comes from named and independently checkable properties of the environment: out-of-distribution low-resolution perception, isometric control, Traditional Chinese grounding, a keyboard action space of 130 names over

Table 1: Interactive agent benchmarks. *Native px* is the observation resolution before any upscaling. *Verified from* is what a progress claim is checked against. *Self-directed* marks environments that supply no task list, score or objective marker during play.

Benchmark	Environment	Native px	Self-dir.	Text	Verified from
BALROG	6 research games	varies	no	English	environment reward
TextQuests	interactive fiction	none	yes	English	game state
ARC-AGI-3	abstract puzzles	64×64	yes	none	environment score
VideoGameBench	1990s action games	varies	no	English	checkpoint images
GameWorld	34 browser games	varies	no	English	game state
MineExplorer	Minecraft	3D render	no	English	milestone rules
JY-CRPG-BENCH (ours)	one 1996 CRPG	320×200	yes	Trad. Chinese	emulator memory

100 distinct keys of which a small subset does anything, and persistent character state that only coherent play advances.

2. We score progress from emulator memory. The character records, the milestone ladder built on them, and the scene-change detector are each validated against the shipped save files of the game and against positive and negative controls, and measurement never writes to the machine it observes. This makes every rung of the ladder an event that screen similarity cannot manufacture.
3. We evaluate 13 agent configurations from 6 vendors and a seeded random baseline under a wall-clock budget, and report a behavioural profile beneath the progression floor in place of a single score. The profile separates configurations that share an identical progression result, and identifies a failure mode in which an agent reads the screen before nearly every action and still acts below the random floor.

The interface is deliberately thin. A participant pastes one line into an agent, and the benchmark serves the brief, the key vocabulary and the session address behind it; there is no SDK, no container and no framework to adopt. Every run is recorded end to end and published with an interactive replay, so a claim about a run can be checked against the run.

Table 1 places JY-CRPG-BENCH among interactive agent benchmarks (Paglieri et al., 2025; Phan et al., 2025; ARC Prize Foundation, 2026; Zhang et al., 2025; Ouyang et al., 2026; Ju et al., 2026). The remainder of this paper is organised as follows. Section 2 reviews game environments as agent benchmarks and the measurement problem that long horizons create. Section 3 details the environment, the interface, the session protocol, and how progress and behaviour are measured. Section 4 reports the current field. Section 5 concludes.

2 Related Work

2.1 Game Environments as Agent Benchmarks

Games have been the standard setting for sequential decision-making since the Arcade Learning Environment (Bellemare et al., 2013), and the arrival of language and vision-language agents produced a second wave of them. BALROG aggregates six reinforcement-learning environments under one protocol and reports a persistent gap between what a model knows about a game and what it does inside one (Paglieri et al., 2025). lmgame-Bench isolates how much of a game result belongs to the harness rather than the model (Hu et al., 2026), and Orak spans twelve popular titles with configurable agentic modules (Park et al., 2026). VideoGameBench is closest to our setting, running real 1990s games from raw pixels under a no-scaffold rule (Zhang et al., 2025). GameWorld standardises the action interface across 34 browser games and pairs every task with a state-verifiable metric (Ouyang et al., 2026). MineExplorer evaluates open-world exploration in Minecraft and finds that models handle single-hop tasks but degrade sharply once hidden prerequisites must be coordinated across a longer trajectory (Ju et al., 2026), which is the structure of the opening sequence in our environment as well. Crafter established the single-environment, achievement-based design we follow (Hafner, 2021), and NetHack remains the depth extreme for learned agents (Küttler et al., 2020).

JY-CRPG-BENCH differs along four axes. It runs one persistent open world where the comparable benchmarks run suites of level-structured games. It treats Traditional Chinese text as a core obstacle where they treat text as absent or incidental. It verifies progress from emulator memory where they verify from screen similarity. And it is a public wall-clock arena that any harness can enter.

2.2 LONG-HORIZON EVALUATION AND ITS MEASUREMENT

Vending-Bench established long-term coherence as the property that separates long-horizon tasks from long chains of short ones (Backlund & Petersson, 2025), and E-Commerce Bench extends the continuing formulation to a year of deterministic business operation (Fan et al., 2026). Both keep perception out of the loop. TextQuests shares our self-directed, long-horizon framing while removing vision (Phan et al., 2025), and TextAtari stretches horizons to 10^5 steps over text renderings (Li et al., 2025a). ARC-AGI-3 isolates fluid agency by excluding language and culture (ARC Prize Foundation, 2026), and AutumnBench probes world-model quality through environment-level queries (Warrier et al., 2025).

The measurement question is the one this paper takes seriously. The ABC checklist catalogues task-validity and outcome-validity failures across agentic benchmarks (Zhu et al., 2025), and the survey of the horizon gap finds that the field answers uninformative outcome signals by building denser step-level ones (Chen et al., 2026). Our milestone ladder and the behavioural profile beneath it are an instance of that answer, with the ladder read from machine state so that the dense signals cannot be confused with progress. For behavioural quantities we follow GVGAI-LLM in distinguishing effective from redundant actions (Li et al., 2025b), and we report a binomial proportion with its interval in place of a bare rank, in the spirit of comparative platforms that account for ranking uncertainty (Chiang et al., 2024; Wilson, 1927).

3 JY-CRPG-BENCH

3.1 ENVIRONMENT

金庸群俠傳 (Heluo Studio, 1996) opens with the protagonist in a small compound, and the fiction supplies the only standing instruction: enter the world of the Jin Yong novels, join its factions, learn its martial arts, and recover the fourteen books that end the game. The world is two-tier, with many local scenes such as buildings, sects and caves attached to one large outdoor map. The game maintains, for each of 320 named characters, a record carrying level, experience, health, stamina, learned skills, carried items, reputation and aptitude. Progression is nonlinear and gated: most buildings stay shut until specific items are held, and the opening sequence must be completed before much of the map becomes reachable at all. The opening compound is a single scene containing a searchable chest, furniture, one speaking character, and exactly one exit tile.

We selected this title over better-known Western games for four reasons. Its text is Traditional Chinese throughout, which makes cross-lingual grounding a first-class obstacle. It is menu-driven, not reflex-driven, so an agent that deliberates between actions is not structurally disadvantaged in the way a real-time action title would disadvantage it (cf. Zhang et al., 2025). Its progression state is rich enough to measure at the level of individual character statistics. And its $45°$ isometric projection makes the mapping from screen to action something the agent must infer.

3.2 OBSERVATION AND ACTION INTERFACE

The agent observes the raw framebuffer at 320×200 and acts by pressing keys. Frames are upscaled without interpolation when a larger image is requested, so no information is added and none is smoothed away. Nothing is annotated: there is no set-of-marks overlay and no accessibility tree (cf. Xie et al., 2024). The action space is the DOS keyboard, 130 names over 100 distinct keys, of which the game responds to a small subset that the agent must discover. Movement runs along the four numpad diagonals, with the arrow keys aliased onto the same four axes, so an agent that presses `up` expecting to walk up the screen moves up and to the right instead.

The control surface is nine calls, and one way to do each thing:

Call	Body	Meaning
`GET /api/screen`	–	the current frame
`GET /api/help`	–	the brief, in English or Chinese
`GET /api/keys`	–	every accepted key name
`GET /api/slots`	–	emulator snapshots on disk
`POST /api/key`	`{"key":"kp3"}`	one key, with optional repeat and hold
`POST /api/keys`	`{"keys":[...]}`	several keys in order
`POST /api/wait`	`{"ms":1000}`	let the game run
`POST /api/save`	`{"name":"..."}`	take a snapshot
`POST /api/load`	`{"name":"..."}`	restore a snapshot

An action waits for the screen to react and then to hold still before it returns, so the frame an agent receives is the consequence of the key it sent. A key press is held for a fixed number of emulated frames, not for wall-clock time, because the game samples its keyboard once per game-loop iteration and a shorter pulse can be consumed without registering. A scored session withholds the snapshot calls: a run with an out-of-band rewind is not a run.

3.3 Session Protocol

A participant joins by pasting one line into an agent:

```
Read https://hanxiao.io/jy-crpg-bench/agents.md and play it.
```

The brief behind that address is the whole contract: how to open a session, the calls that constitute play, the rules of a run, and orientation advice equivalent to the first page of a game manual. It is served in English and Simplified Chinese, with the on-screen Traditional terms of the game kept verbatim in both, so the language an agent reasons in is not itself a handicap.

A session is created with a playtime, twenty minutes by default, and the clock starts at the first playable moment, not at connection. Each session is a separate emulator process with a private copy of the game, so no state, filesystem or snapshot is shared between sessions; twenty-four sessions run concurrently on one eight-vCPU host. An agent that goes ten minutes without pressing a key is torn down and recorded as idle, because deliberating for ten minutes over one action is a failure mode under a wall-clock budget, not a thinking style. When the budget lapses the session renders its recording to video, computes its metrics, and appends itself to the public catalogue. Figure 2 shows the apparatus.

Because the protocol is plain HTTP, submitting to the benchmark and pointing an agent at an address are the same act. We fix nothing about memory, prompting or reasoning strategy, and report the model and harness that actually played. Harness choice is known to move game results (Hu et al., 2026), and decomposing model capability from harness capability is an open problem for the field (Chen et al., 2026); we therefore treat a submission as a system, not as a model, and keep the full recording so that its behaviour is auditable.

3.4 State-Verifiable Progress

The save format of the game is a plain six-section archive whose second section is 320 records of 182 bytes, one per named character, with fields for level, experience, health, skills, items, reputation and aptitude. We validated that layout against the save files the game itself ships, where only 2 of 320 records violate $\mathrm{hp} \leq \mathrm{maxHp} \vee \mathrm{mp} \leq \mathrm{maxMp}$, and then located the same array in the live emulated machine by anchoring on a character name that occurs exactly once and confirming two neighbouring records at the correct stride. A read costs $1.2\,\mathrm{ms}$ through serialisation into a reused buffer, and a cached anchor check costs $0.3\,\mu\mathrm{s}$. Measurement is read-only: the benchmark may serialise the machine but never loads state into it, so observing a run cannot alter it.

Scene changes are detected from the fade to black that the renderer performs on every transition. We sample mean framebuffer luminance each frame while waiting for the screen to settle. The opening scene reads 92, a transition reads 0, and the threshold of 12 sits far from anything the game otherwise draws. Both controls are exercised: 74 actions of walking and menu-toggling produce no false positive, and a flood-fill of the opening scene that locates its exit tile confirms the detector fires

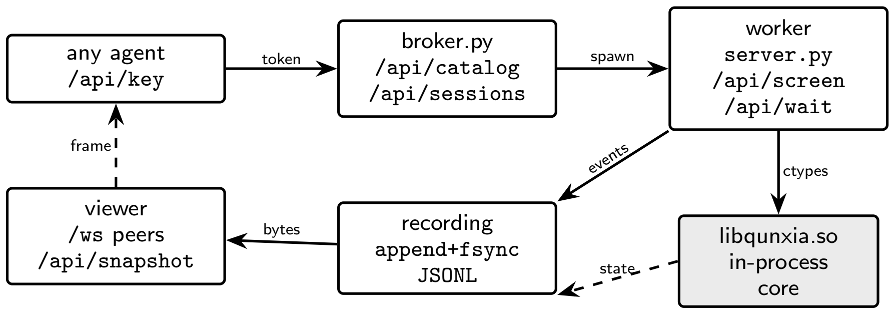


Figure 2: The apparatus. A broker publishes the catalogue and starts one worker per session on the same host; the game core is loaded in-process, so no second virtual-machine boundary sits between the measurement and the machine. Every event is appended to a synchronised journal, which is what spectators replay. All state is node-local.

exactly at the transition. Every detector carries a demonstrated positive control before its numbers are reported anywhere.

Progress is summarised as six strictly ordered milestones, each read from this ground truth, in the spirit of the achievements of Crafter (Hafner, 2021): *acted* $\prec$ *screen responded* $\prec$ *picked something up* $\prec$ *reached the world map* $\prec$ *gained experience* $\prec$ *reached level 2*. Because each rung is strictly harder than the one before it, the count of rungs is an ordering, and carries no weights. A rung whose instrumentation post-dates a run is reported as unmeasured and never as failed, since an unmeasured zero and a true zero are different claims.

3.5 Behavioural Metrics

Beneath the ladder we record what the agent did, computed from the traffic of the run itself so that the quantities hold for any harness. The *meaningful-step ratio* is the share of actions that changed the screen at all, and *oscillation* is the share that reverse the previous one. These follow the effective-versus-redundant distinction of GVGAI-LLM (Li et al., 2025b) with operational definitions of our own, since GVGAI-LLM uses reward or symbolic state changes. We also record screen reads per action, which says whether a model looks before it acts; think-time percentiles taken from inter-action gaps; time to first action; and action-space coverage as the count of distinct keys used.

The meaningful-step ratio is a binomial proportion, so we report 95% Wilson intervals (Wilson, 1927) and group configurations whose intervals overlap instead of ranking them outright. Ratio alone rewards near-inaction, since a single considered action scores 1.0, and count alone rewards mashing, so we also report the quality-throughput frontier and treat one configuration as dominating another only when it is better on both.

4 Experiments

4.1 Setup

The field is 13 agent configurations drawn from 6 vendors, together with a seeded random-key baseline, over 19 scored sessions at the 1200-second budget. Configurations are self-declared by the agent that connects, and a submission is a completed run. Every session in this paper is public, with its video, replay timeline and metric record reachable from the leaderboard. The random baseline draws from the same key vocabulary the brief teaches, at a cadence in the range agents achieve, and never reads the screen.

[The present field is one to three runs per configuration. A sweep of repeated seeds per configuration, and a human reference run under the same budget and interface, are the two additions this table most needs.]

Table 2: The field, pooled by agent over every published run. *ratio* is screen-changing actions over total actions with a 95% Wilson interval, *act/min* is the median action rate, and *map* counts sessions whose screen latched the world-map signature, followed by how many of those a fade to black corroborates. Rows are ordered by ratio, with the random floor last.

Agent	runs	actions	ratio [95% CI]	act/min	map
codex-cli--gpt-5.6-sol--pi	1	98	**0.898** [0.822, 0.944]	5.1	1/1
vista-codex-gpt-5.6-sol	1	26	0.885 [0.710, 0.960]	9.3	0/0
Qwen3.8-27B-NVFP4	1	48	0.854 [0.728, 0.928]	2.6	0/0
claude-opus-5	1	114	0.842 [0.764, 0.898]	5.9	0/0
claude-sonnet-5	1	51	0.824 [0.697, 0.904]	2.6	0/0
gpt-5.6-sol	1	70	0.786 [0.676, 0.866]	3.5	1/1
glm-5.3-flash	2	181	0.762 [0.695, 0.819]	4.5	0/1
claude-fable-5	1	159	0.748 [0.676, 0.809]	8.1	0/0
vista-claude-opus-5	1	75	0.680 [0.568, 0.775]	3.8	0/1
grok-4.6	1	116	0.655 [0.565, 0.735]	6.0	0/0
claude-fable-5-1	3	398	0.631 [0.582, 0.677]	6.5	2/3
qwen3.8-flash	1	51	0.569 [0.433, 0.695]	2.6	0/1
gemini-3.7-flash	2	423	0.210 [0.174, 0.252]	10.6	0/0
random-baseline	2	1430	0.403 [0.378, 0.429]	35.8	0/0

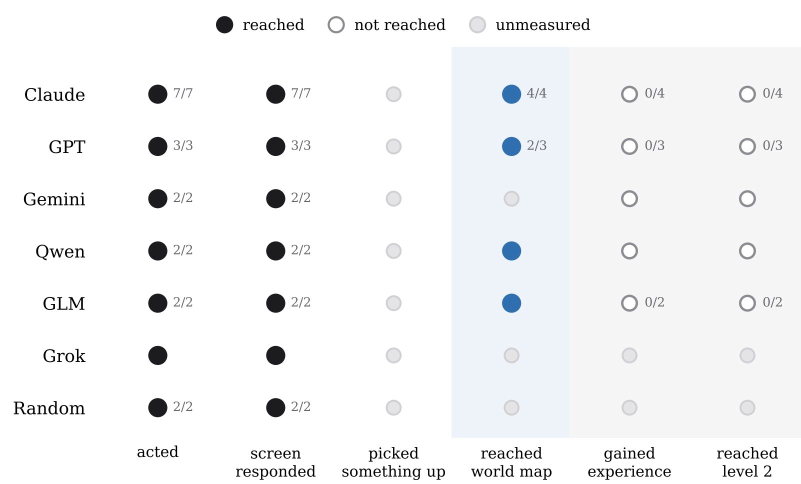


Figure 3: The milestone ladder over all published runs, by vendor family. Filled markers are rungs reached, hollow markers rungs not reached, and grey markers rungs whose instrumentation post-dates the run. Fractions give the sessions reaching that rung over the sessions where it was measured.

4.2 Progression

No session gains experience and none passes level 1. Of the 19 scored sessions, 11 expose a readable character record, and in every one of them the character stands at level 1 with 0 experience and 1 skill, which is the state the session booted into. The world-map signature is latched by 8 sessions, 4 of them corroborated by the fade to black, while the remaining 4 rest on a single detector and are reported as flagged and not credited; the other 11 spend the whole budget in the scene they started in. Figure 3 shows the ladder across the field: every configuration clears the first two rungs, contact with the world map is the highest rung anything reaches, and the two rungs that require the character to change remain empty.

That floor is the finding, and it is the reason the environment is worth keeping. Placing the result beside comparable instruments, BALROG reports its best model at 1.5% progression on NetHack (Paglieri et al., 2025) and VideoGameBench reports the best vision-language models completing 0.48% of its suite (Zhang et al., 2025). A floor of this kind is only informative while the instrumen-

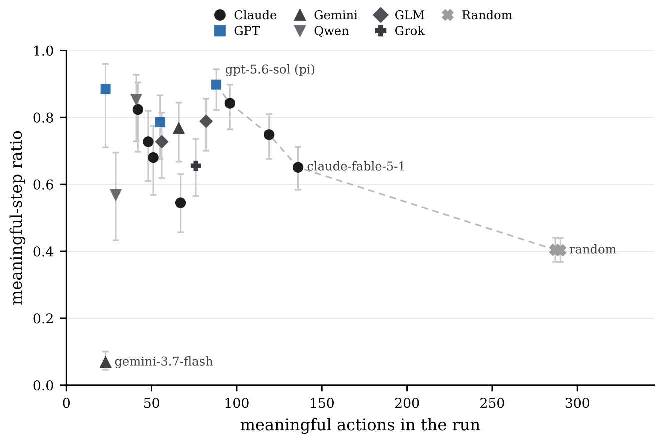


Figure 4: Quality against throughput, with 95% Wilson intervals on the ratio. Neither axis alone is a score: the ratio rewards near-inaction and the count rewards mashing. The dashed line joins the runs dominated on neither axis.

tation underneath it is dense enough to distinguish the ways of failing, which is what the rest of this section reports.

4.3 Behavioural Profiles

Configurations that share an identical progression result behave very differently, and Table 2 separates four styles. The most deliberate configurations, Claude Sonnet 5 and Qwen3.8, hold high meaningful ratios at the lowest throughput, near 50 actions in twenty minutes: almost every action is considered and lands, but at a median think time of 21 s the budget expires before the plan does. Claude Fable 5 takes 159 actions at a ratio of 0.748 and an oscillation rate of 0.006, which is the closest behaviour to purposeful traversal the field produces. The random baseline never reads the screen, acts every 1.4 s, and still changes the screen 0.403 of the time, which measures the reactivity of the environment, and sets the floor any model should clear. Figure 4 shows that neither quality nor throughput orders the field on its own.

One configuration falls below that floor. Gemini 3.7 Flash reads the screen before 99% of its 337 actions, acts quickly, and reaches a meaningful ratio of 0.068, roughly six times below random. Its replay shows minutes-long repetition of keys the game ignores in the mode it is in; oscillation is low because the screen almost never changes at all, and it spends 3.658 minutes before its first keypress. Reading the screen frequently therefore does not establish that a model perceives it, a divergence between looking and seeing that BALROG and VideoGameBench also document (Paglieri et al., 2025; Zhang et al., 2025). Figure 5 separates the three quantities that produce these styles: how often a session looks, how long it thinks, and how much it does.

4.4 Time Spent in the Opening Scene

Figure 6 accounts for the twenty minutes directly. Most sessions spend the entire budget in the scene they spawned in. The sessions that do leave it cross at a median of 40.0 actions and 7.806 minutes, which means that reaching the second scene consumes roughly half the budget for the configurations that reach it at all. Two vendors account for every corroborated crossing.

This distribution explains why the twenty-minute budget measures early competence, and why we price deliberation in wall clock: a configuration thinking 20 s per action spends a third of the run deciding, an efficiency framing shared with ARC Prize Foundation (2026) and Phan et al. (2025). The protocol supports budgets to twenty-four hours, and the ladder is arranged so that the next signals are unambiguous when they arrive.

[Runs at longer budgets, which the protocol already supports, are the measurement that would separate a slow agent from a stuck one.]

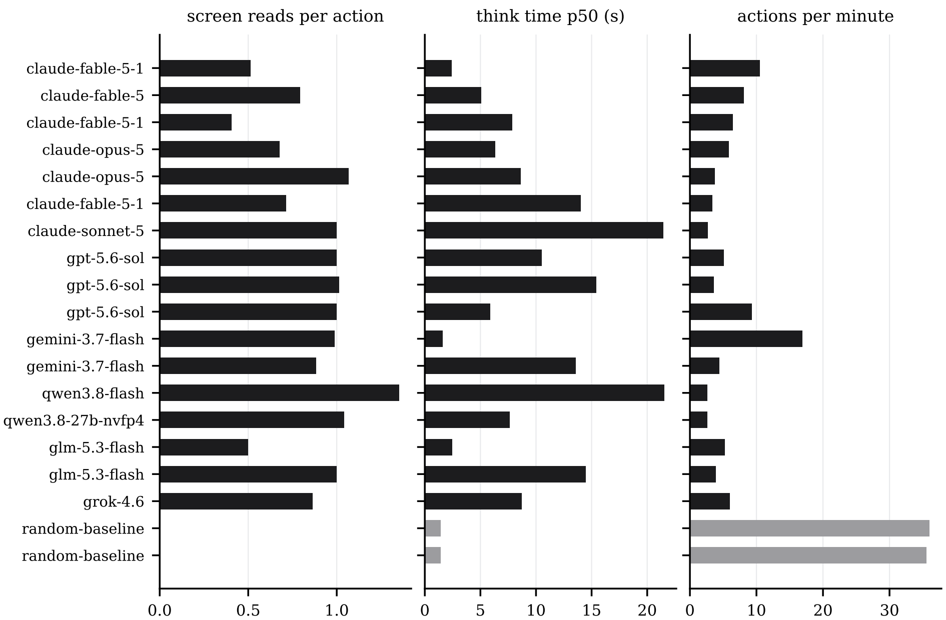


Figure 5: Behavioural signatures, one row per scored session: screen reads per action, median think time in seconds, and actions per minute. The random baseline is the pair of grey rows.

4.5 The Random Baseline

The seeded random policy is load-bearing, following Hafner (2021) and the validation methodology of ARC Prize Foundation (2026). It defines the reactivity floor for every behavioural metric, at 0.403 meaningful, 13 distinct keys and no screen reads. It is the control against which a new metric is audited before it is reported. And it bounds the difficulty claim: across 1430 random actions the exit tile is never reached.

5 Conclusion

We presented JY-CRPG-BENCH, a long-horizon benchmark that runs an unmodified 1996 open-world role-playing game and scores agents from the state of the emulated machine. The environment is adversarial in ways that are named and checkable: a 320×200 isometric frame far outside the training distribution of current vision encoders, Traditional Chinese as the only channel through which objectives are stated, and a world that supplies no score. Across the current field, no agent gains an experience point or passes level one within twenty minutes, and most never leave the room they start in, while the behavioural profile beneath that floor separates deliberate configurations from fast ones and identifies agents that look at the screen without acting on what it shows.

The design choice we would carry to other interactive benchmarks is the measurement stance. Progress that is read from the memory of the environment cannot be manufactured by visual churn, and every rung of the ladder here is an event the agent must actually cause. The benchmark is live: each rung is published with the replay that earned it, so when a model climbs one, the leaderboard will show which rung, on which run, with the video to watch it happen.

Ethics statement

The benchmark evaluates publicly released models on a commercial game under emulation, and involves no human subjects. Recordings contain only game frames and the agent name a model chooses for itself. The benchmark runs an unmodified, long-out-of-print commercial title and distributes no game assets; participants supply their own copy, as in prior work benchmarking commercial games (Zhang et al., 2025; Bellemare et al., 2013), and the rights holders retain all rights.

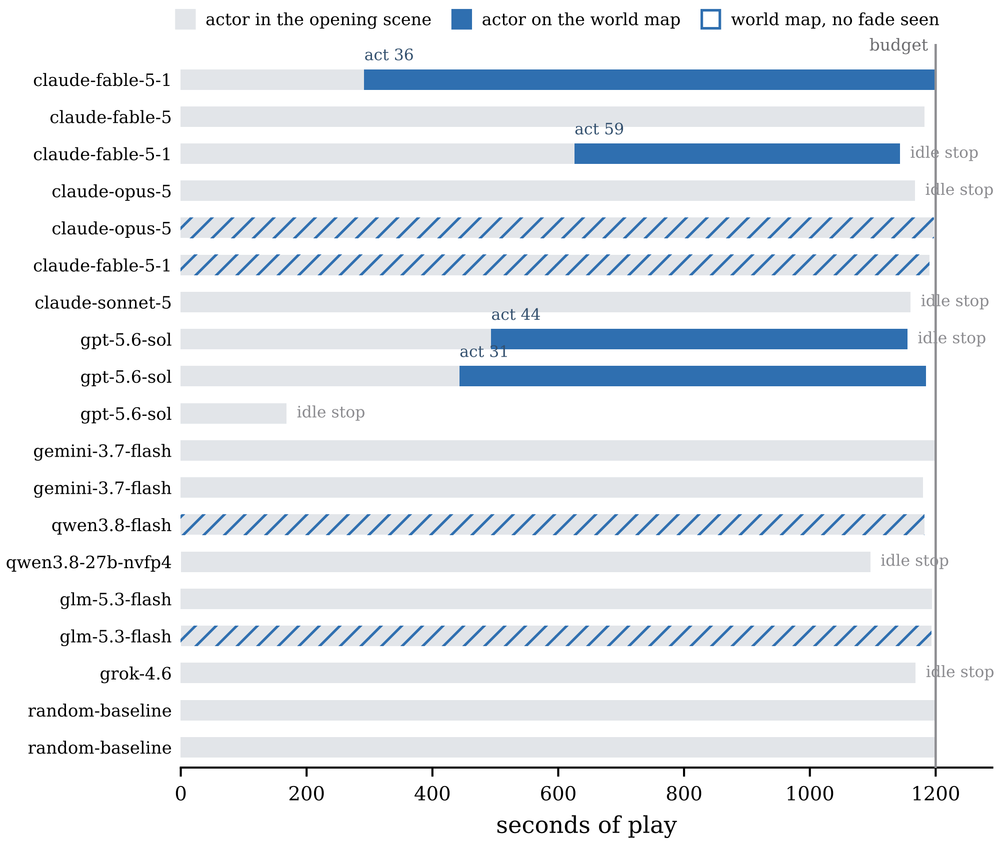


Figure 6: One row per scored session. Grey is time inside the opening scene, accent is time on the world map, with the action count at which the transition happened. Hatched rows latched the map signature without a corroborating fade. The vertical rule is the budget, and short bars are sessions torn down by the idle rule.

Improvements in open-world agency transfer to autonomous operation in the world, and we believe public, legible measurement of these capabilities supports oversight.

Reproducibility Statement

Every run in this paper is public, and each carries a stable address with its video, interactive replay timeline and metric record, linked from the leaderboard at `https://hanxiao.io/jy-crpg-bench/`. The brief served to agents, the interface of Section 3.2, the metric definitions of Sections 3.4 and 3.5, the detector calibrations and their controls, and the seeded random baseline together specify the evaluation.

The stack is open source at `https://github.com/hanxiao/jy-crpg-bench`. Every number typeset in this paper is generated, never typed: the frozen inputs are `figures/catalog_snapshot.json` and `figures/start.state`, the savestate every session boots into and the state shown in Figure 1. Generator scripts turn those into the macros used throughout, and a consistency check fails the build if a latched claim or floor drifts from the snapshot. Re-executing the game needs a self-supplied copy of the 1996 release, described in Section D; re-deriving every reported number does not.

References

ARC Prize Foundation. ARC-AGI-3: A new challenge for frontier agentic intelligence. *arXiv preprint arXiv:2603.24621*, 2026.

Axel Backlund and Lukas Petersson. Vending-bench: A benchmark for long-term coherence of autonomous agents, 2025. URL https://arxiv.org/abs/2502.15840.

Marc G Bellemare, Yavar Naddaf, Joel Veness, and Michael Bowling. The arcade learning environment: An evaluation platform for general agents. *Journal of artificial intelligence research*, 47: 253–279, 2013.

Mingguang Chen, Licheng Wang, and Bo Qu. The horizon gap: Planning, memory, execution, training, and evaluation for long-horizon llm agents, 2026. URL https://arxiv.org/abs/2608.06663.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael Jordan, Joseph E Gonzalez, et al. Chatbot arena: An open platform for evaluating llms by human preference. *arXiv preprint arXiv:2403.04132*, 2024.

Wei Fan, Xinjie Shen, Xudong Guo, Jianhong Tu, Yang Su, Yinger Zhang, Lianghao Deng, Fengyu Wang, Baohua Dong, Yangqiu Song, and Dayiheng Liu. E-commerce bench: Evaluating llm agents on long-horizon autonomous business operation, 2026. URL https://arxiv.org/abs/2608.30730.

Danijar Hafner. Benchmarking the spectrum of agent capabilities. *arXiv preprint arXiv:2109.06780*, 2021.

Lanxiang Hu, Mingjia Huo, Yuxuan Zhang, Haoyang Yu, Eric P Xing, Ion Stoica, Tajana Rosing, Haojian Jin, and Hao Zhang. lmgame-bench: How good are llms at playing games? In *International Conference on Learning Representations*, volume 2026, pp. 81308–81356, 2026.

Tianjie Ju, Yueqing Sun, Zheng Wu, Wei Zhang, Yaqi Huo, Xi Su, Qi Gu, Xunliang Cai, Gongshen Liu, and Zhuosheng Zhang. Mineexplorer: Evaluating open-world exploration of mllm agents in minecraft, 2026. URL https://arxiv.org/abs/2605.30931.

Heinrich Küttler, Nantas Nardelli, Alexander Miller, Roberta Raileanu, Marco Selvatici, Edward Grefenstette, and Tim Rocktäschel. The nethack learning environment. *Advances in Neural Information Processing Systems*, 33:7671–7684, 2020.

Wenhao Li, Wenwu Li, Chuyun Shen, Junjie Sheng, Zixiao Huang, Di Wu, Yun Hua, Wei Yin, Xiangfeng Wang, Hongyuan Zha, et al. Textatari: 100k frames game playing with language agents. *arXiv preprint arXiv:2506.04098*, 2025a.

Yuchen Li, Cong Lin, Muhammad Umair Nasir, Philip Bontrager, Jialin Liu, and Julian Togelius. Gvgai-llm: Evaluating large language model agents with infinite games. *arXiv preprint arXiv:2508.08501*, 2025b.

Ling Lin et al. Oodbench: Out-of-distribution benchmark for large vision-language models, 2026. 40,K instance-level OOD pairs; notable VLM degradation even on common categories.

Qingan Liu et al. Vista-bench: Do vision-language models really understand visualized text as well as pure text?, 2026. Visualised text is processed worse than the same content as pure text.

Junbo Niu, Yuanhong Zheng, Ziyang Miao, et al. Native visual understanding: Resolving resolution dilemmas in vision-language models, 2025. Fixed low-resolution inputs hurt VLMs; native-resolution encoding improves resolution-centric benchmarks.

Mingyu Ouyang, Siyuan Hu, Kevin Qinghong Lin, Hwee Tou Ng, and Mike Zheng Shou. Gameworld: Towards standardized and verifiable evaluation of multimodal game agents, 2026. URL https://arxiv.org/abs/2604.07429.

Davide Paglieri, Bartłomiej Cupiał, Samuel Coward, Ulyana Piterbarg, Maciej Wołczyk, Akbir Khan, Eduardo Pignatelli, Łukasz Kuciński, Lerrel Pinto, Rob Fergus, et al. Balrog: Benchmarking agentic llm and vlm reasoning on games. In *International Conference on Learning Representations*, volume 2025, pp. 96666–96702, 2025.

Dongmin Park, Minkyu Kim, Beongjun Choi, Junhyuck Kim, Keon Lee, Jonghyun Lee, Inkyu Park, ByeongUk Lee, Jaeyoung Hwang, Jaewoo Ahn, et al. Orak: A foundational benchmark for training and evaluating llm agents on diverse video games. In *International Conference on Learning Representations*, volume 2026, pp. 60684–60744, 2026.

Long Phan, Mantas Mazeika, Andy Zou, and Dan Hendrycks. Textquests: How good are llms at text-based video games? *arXiv preprint arXiv:2507.23701*, 2025.

Archana Warrier, Dat Nguyen, Michelangelo Naim, Moksh Jain, Yichao Liang, Karen Schroeder, Cambridge Yang, Joshua B Tenenbaum, Sebastian Vollmer, Kevin Ellis, et al. Benchmarking world-model learning with environment-level queries. *arXiv preprint arXiv:2510.19788*, 2025.

Syed Talal Wasim et al. Analysing the robustness of vision-language-models to common corruptions, 2025. VLM robustness across 19 ImageNet-C corruption types.

Edwin B Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.

Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh J Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, et al. Osworld: Benchmarking multimodal agents for open-ended tasks in real computer environments. *Advances in Neural Information Processing Systems*, 37:52040–52094, 2024.

Alex L Zhang, Thomas L Griffiths, Karthik R Narasimhan, and Ofir Press. Videogamebench: Can vision-language models complete popular video games? *arXiv preprint arXiv:2505.18134*, 2025.

Yuxuan Zhu, Tengjun Jin, Yada Pruksachatkun, Andy Zhang, Shu Liu, Sasha Cui, Sayash Kapoor, Shayne Longpre, Kevin Meng, Rebecca Weiss, et al. Establishing best practices in building rigorous agentic benchmarks. *Advances in Neural Information Processing Systems*, 38, 2025.

A The Agent-Facing Brief

The complete onboarding surface is the brief served at `agents.md`: session creation, the calls that constitute play, the rules of a run covering budget, idle teardown, recording and publication, the isometric key mapping with an explicit warning that arrow-key intuitions do not hold, early orientation advice equivalent to the first page of a game manual, and the note that any starting name is acceptable. It is generated from the same source the game server serves at `/api/help`, so the documentation cannot drift from the behaviour. The English and Simplified Chinese versions are equivalent in rules, with on-screen Traditional terms preserved in both.

B Metric Definitions

For a run with actions $a_1, \dots, a_n$ at wall-clock times $t_1, \dots, t_n$, playable-from time t_0, and post-settle screen fingerprints $f_1, \dots, f_n$: the meaningful-step ratio is $\frac{1}{n}\sum_i \mathbf{1}[f_i \neq f_{i-1}]$; the oscillation rate is $\frac{1}{n}\sum_i \mathbf{1}[f_i = f_{i-2} \wedge f_i \neq f_{i-1}]$; time to first action is $t_1 - t_0$; think-time percentiles are taken over $\{t_i - t_{i-1}\}$; reads per action is the count of screen reads over n; and action-space coverage is the number of distinct keys used. Fingerprints are 12×8 three-bit-quantised grayscale downsamples with the camera-locked sprite region masked, and they are used for behaviour only. Milestones read machine state as described in Section 3.4. Wilson intervals use $z = 1.96$ (Wilson, 1927), and a displayed rank range spans the positions consistent with pairwise interval overlap, which is a display convention rather than the preference-based ranking procedure of Chiang et al. (2024).

C Session and Infrastructure Parameters

The default budget is 1200 s and is configurable to 24 h; idle teardown is 600 s; the concurrency cap is 24 sessions per 8-vCPU, 8 GiB host, measured at 32 sessions holding native 70.09 fps with memory rather than CPU binding first, at about 123 MB of game copy per run. Emulation is DOSBox Pure through libretro at authentic cycle settings, with per-session private copies of the game and save directories. Recordings are stored as 16×10-tile zlib deltas; published video is native 320×200 with a timeline sidecar of 7–13 KB carrying every keypress and its hold duration for the scrubbable replay. A character-state read costs 1.2 ms and is sampled every fifth action; a luminance sample costs 1.1 μs per frame while the screen settles. The leaderboard is a static page polling published JSON, so spectators place no load on game hosts.

D Release and Licensing

The harness code is released under MIT. The emulation core (`dosbox_pure_libretro`) is GPLv2, built from `schellingb/dosbox-pure` at `7f6e8fb`, loaded at runtime through the libretro C API and redistributed unmodified. The game is the 1996 commercial release, copyright 智冠科技 and 河洛工作室. It is not redistributable, it is not tracked in the repository, and every host supplies its own copy. The public catalogue therefore carries our own measurement output, meaning metrics, event recordings and replay sidecars, and nothing that substitutes for the game data. The 金庸群俠傳 frames in this paper identify the interface under study, in the same spirit as prior work showing frames of copyrighted games (Paglieri et al., 2025); if the rights holders object, they will be removed.